

GenAI-Driven Hyperlocal Operations & CX Architecture

Operational Empathy Framework, Multimodal Dispute Triage, & Kitchen NLP Disambiguation

Document Status

Approved for Execution

Target Domain

FoodTech / Quick-Commerce

Implementation

4-Quarter Rollout

Target Impact

78% Dispute TAT Reduction

1. Executive Summary & Problem Statement

Operational Friction

Conventional analytics (SQL/Amplitude funnels) isolate *what* failed (e.g., delivery time exceeding SLA by 12 mins), but lack root-cause visibility into frontline failure modes: merchant thermal printer jams, non-vented packaging steam condensation, and multi-tenant high-rise navigation bottlenecks.

Product Solution

A unified AI product system combining a mandatory 5-day Frontline PM Rotation Model with four AI microservices: Voice-to-Jira Ingestion, Multimodal Vision Dispute Triage, KDS Instruction Normalization, and Telemetry-Aware Conversational Ordering.

OPERATIONAL PILLAR	LEGACY BASELINE	PRODUCTION TARGET	BUSINESS KPI IMPACT
Customer Meal Discovery	25 nested filters; 15 min session	Contextual conversational prompt	-60% Time-to-Cart +28% Checkout Conv.
Order Dispute Resolution	12.5 min manual agent chat	0.4s Multimodal Vision verification	-78% Cycle Time (24s avg handle)
Cloud Kitchen Assembly	Unstructured text on paper tickets	KDS NLP color-coded visual icons	-42% Customization Defect Rate
Field Immersion Ingestion	Manual unstructured post-mortems	Voice-to-PRD structured epic generation	5x Velocity in Backlog Prioritization

2. Operational Protocol: The Frontline Rotation Model

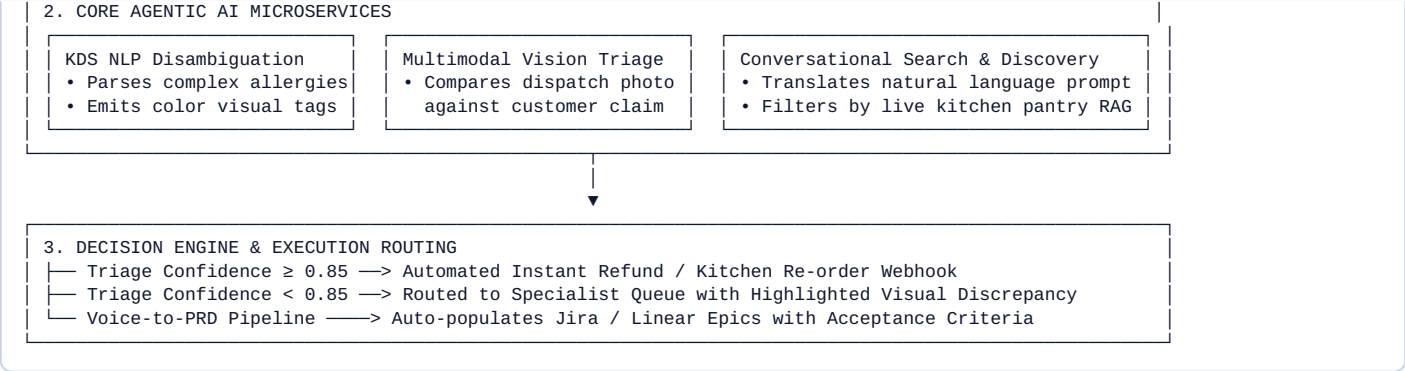
To ground engineering roadmaps in frontline reality, every Product Manager, Tech Lead, and Product Designer completes a mandatory quarterly rotation schedule:

ROTATION TRACK	OPERATIONAL FOCUS	IDENTIFIED FAILURE MODES	AI / ENGINEERING DELIVERABLE
Days 1 & 2: Kitchen & Packing	Standing on line, bagging under heat lamps, dispatch validation.	Steam vent blockage degrading food texture; receipt printer buffer overflow during rush hours.	Automated packaging dynamic rules & KDS visual order parsing.
Days 3 & 4: Courier Ride-Along	Last-mile co-pilot with delivery fleet partners.	Service elevator bottlenecks; missing gated security pin codes; parking friction.	Crowdsourced micro-navigation & automated gate-code extraction.
Day 5: Live Support Triage	Direct handling of tier-1 escalations and defect claims.	False-positive spill claims; agent fatigue; slow manual photo validation.	Multimodal Vision AI for automated instant refunds.

3. End-to-End System Architecture & Data Pipeline

1. INGESTION & FRONTLINE TELEMETRY LAYER

- PM Voice Memos: Whisper ASR → Audio-to-Text Transcription Gateway
- Merchant Telemetry: Live Pantry Inventory, Expiry Dates, Prep Queue Depth via POS Webhooks
- Dispatch Snapshots: High-Res Edge Cameras at Merchant Dispatch & User Uploaded Dispute Photos



Q2	Multimodal Dispute Triage	Integrate dispatch cameras; train defect vision models; launch automated refund webhooks.	Dispute handle time reduced from 12.5m to < 60s; 70% automated resolution.
Q3	KDS NLP & Pantry Balancing	Deploy ticket normalization engine; connect real-time ERP inventory to merchant dynamic menu builder.	40% reduction in customization errors; 18% reduction in raw produce waste.
Q4	Conversational Discovery	Launch natural language ordering agent with real-time pantry RAG integration across mobile apps.	25%+ increase in search-to-cart conversion; 60% drop in session duration.